RITVIK MAHAPATRA

Fresno, CA | 559-321-6628 | ritvik.mahapatra@yahoo.com | LinkedIn | GitHub | Google Scholar

Summary

AI Engineer with hands-on experience shipping end-to-end ML systems, from real-time computer vision (pose estimation) to NLP classification pipelines, using Python, PyTorch, TensorFlow, and scikit-learn, deployed to production on GCP with FastAPI and Docker. Published researcher in applied ML for quality assessment and vision systems.

EXPERIENCE

Dept. of Computer Science, Fresno State | ***Teaching Associate*** **Aug 2024 - May 2026**

- Led weekly lab sessions across Computer Architecture, Data Structures, and introductory C++ programming, translating lecture concepts into hands-on exercises that reinforced core course material for students struggling to apply theory independently.
- Guided 40+ students through assignments each semester, troubleshooting and debugging scripts to resolve technical blockers in real time, which increased on-time assignment submission rates.

LTIMindtree | ***Data Engineer Trainee Intern*** **Feb 2023 - Oct 2023**

- Completed a rigorous, assessment-gated training program spanning Java, Python, SQL, and cloud services, achieving a 100% pass rate across 10 weekly technical evaluations.
- Applied MongoDB, Hadoop, PySpark, and Hive in hands-on lab exercises to practice ingesting, processing, and querying large-scale datasets, developing practical familiarity with a full big-data pipeline stack.
- Gained hands-on exposure to core AWS and Azure services and practiced writing SQL queries against relational datasets in lab exercises, building foundational proficiency in cloud platforms and data querying.

PUBLICATIONS

- R. Mahapatra, A. S. Reen, K. Borkar.IMVS: Interactive Medical Volume Segmentation with Test-Time Adaptation - A New Method for Annotating Radiology Datasets.*MICCAI HAIC Workshop, 2026 (accepted, pending publication)*
- S. Rahul, R. Mahapatra, K. Gaurav.Brief Review of Deep Learning Techniques Employed in Face Mask Classification.*AIMLA, 2024. DOI · Cited by 2*
- R. Mahapatra, K. Gaurav.Real Time Quality Assessment in a Production Line: Machine Learning Approach.*MoSICom, 2023. DOI · Cited by 2*

PROJECTS

LiftFormVision: AI-Powered Exercise Form Analysis App | GitHub **Jan 2026 - Aug 2026**

- Created a computer-vision system for real-time form-checking across 2 exercises (squat, overhead press), building a dedicated isolated pipeline for each.
- Squat depth-error detector reached 0.846 F1 on Fitness-AQA; OHP uses a temporal-CNN knee-error classifier.
- Hosted the app locally via Flask, supporting both video upload and live-camera capture for real-time feedback.

Full-Stack NLP Sentence Classification System | GitHub **Jun 2026 - Jul 2026**

- Built a 4-class oppression-type classifier (embeddings + SVD + logistic regression) behind a FastAPI backend, reaching 77.8% accuracy and 96.3% top-2 accuracy on held-out test data.
- At a 0.70 confidence threshold, 72.8% of predictions are handled directly at 88.98% accuracy, with the rest routed to an LLM fallback; closed-loop retraining on GCP reserves 15% of feedback in a frozen eval pool.
- Trained on ~750 LLM-generated synthetic sentences per label, deployed on Google Cloud Run with GCS-based model hot-reloading so retraining can promote new models without a redeploy.

Virtual Reality Object Manipulation Environment | GitHub **Oct 2025 - Dec 2025**

- Built the shared Unity/XR environment and infrastructure (persistent XR rig, event-driven feedback, object-manipulation framework) supporting 6 HCI interaction features.
- Used modular additive-scene architecture to separate environment from interaction logic; validated through user testing with 10+ participants.

EDUCATION

California State University, Fresno **Jan 2024 - Aug 2026**
Master of Science, Computer Science (GPA: 3.87/4.00)
- **Achievements:** Academic All-Star Award recipient

Manipal University, Jaipur **Jul 2019 - Jun 2023**
Bachelor of Technology, Mechatronics Engineering (GPA: 3.52/4.00)
- **Achievements:** Dean's List (7 semesters), Excellence in Academics

TECHNICAL SKILLS

- **Programming Languages**: C, C++, C#, Python, Java, SQL
- **ML / DL Frameworks**: PyTorch, TensorFlow/Keras, NumPy, scikit-learn, Hugging Face Transformers, Sentence-Transformers, MediaPipe, OpenCV
- **Backend and Deployment**: FastAPI, Flask, Docker, Groq API
- **Cloud and MLOps**: Google Cloud Platform (Cloud Run, Cloud Build, Firestore, Cloud Storage, Cloud Scheduler), AWS, Azure, CI/CD, MLOps
- **Data & Databases**: MySQL, MongoDB, PySpark, Hive, Hadoop HDFS
- **AI & ML Concepts**: Machine Learning, Deep Learning, Natural Language Processing, Computer Vision, Vision Transformers, Generative AI, LLMs
- **Other Tools**: Unity, XR Interaction Toolkit, OpenXR, Git, MATLAB

EXTRACURRICULARS

OpenAI | ***Campus Ambassador*** **Aug 2025 - Dec 2025**

- Led initiatives as an OpenAI Campus Ambassador to increase student awareness and understanding of AI tools and applications at CSU Fresno.

Electronics Society | ***Senior R&D Coordinator*** **Aug 2021 - Jun 2022**

- Engaged in research initiatives in electronics and robotics, contributing to undergraduate projects and knowledge sharing.